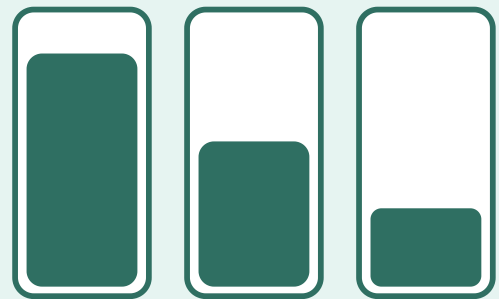


228 cells

Calendar, cyclic, and profile ageing
0–40 °C

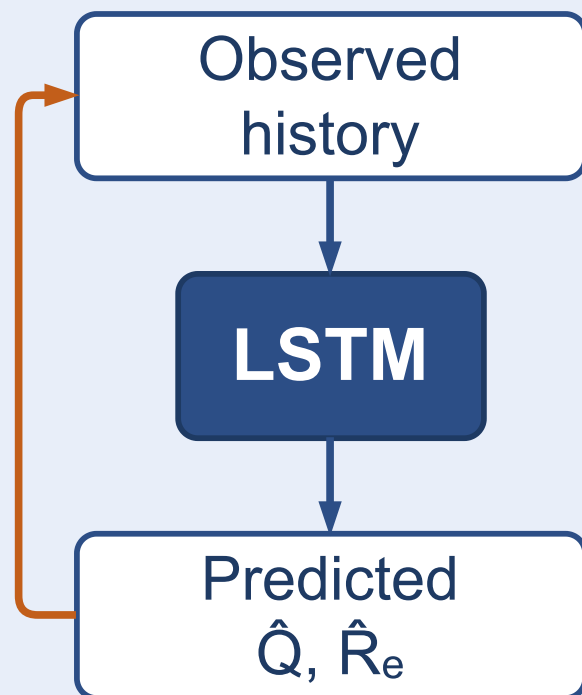


Capacity ↓
Resistance ↑

3,980
checkups

Recursive rollout

No future health measurements



Predictions feed the next step

Matched multi-objective search

15 runs × 280 evaluations per method

NSGA-II

NSGA-III

Random

validation macro MAPE ↓
parameters ↓ · latency ↓

Final mean hypervolume

NSGA-II 1.1972 vs random 1.1410

Mean evaluations to reach
random-search final mean HV

NSGA-II 71

Random 167

Reached in 15/15 runs vs 8/15

Selected model

92.5%

fewer parameters
486,978 → 36,418

38.9%

lower forward latency
0.696 → 0.425 ms

**No significant
accuracy difference**

1.810% vs 1.856% MAPE